Number Systems

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Remark

Remark. Probability and statistics count with the integers and natural numbers and measure with the real numbers. This page defines those number systems. It also defines the words that sort their members: even and odd integers, and rational and irrational real numbers.

Natural numbers

Exercise 0.1 (Which numbers are natural?)**.** List the elements of the set $\{n \in \mathbb{N} : n < 3\}$. Is your answer the same in every textbook?

Solution

Solution 0.1. The answer depends on whether the source counts 0 as a natural number:

- if $\mathbb{N}$ starts at 0, the set is $\{0, 1, 2\}$;
- if $\mathbb{N}$ starts at 1, the set is $\{1, 2\}$.

Both conventions are in common use, so the answer is not the same in every textbook.

Definition 0.1 (Integers)**.** The **integers** (Wikipedia contributors 2026a) are the whole numbers, together with their negatives and 0:

$$\mathbb{Z} \stackrel{\text{def}}{=} \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

An **integer** is an element of $\mathbb{Z}$.

Definition 0.2 (Positive integers)**.** The integers (Definition 0.1) greater than 0 are the **positive integers**.

Definition 0.3 (Negative integers)**.** The integers (Definition 0.1) less than 0 are the **negative integers**.

Example 0.1 (Integers and non-integers)**.**

- -7, 0, and 12 are integers.
- $\frac{1}{2}$ and -2.5 are not integers.
- $\frac{6}{3}$ is an integer, because $\frac{6}{3} = 2$.

Definition 0.4 (Even integer)**.** An integer (Definition 0.1) n is **even** (Wikipedia contributors 2026d) if $n = 2k$ for some integer k.

Definition 0.5 (Odd integer)**.** An integer (Definition 0.1) n is **odd** (Wikipedia contributors 2026d) if $n = 2k + 1$ for some integer k.

Example 0.2 (Even and odd integers)**.**

- 6 is even, because $6 = 2 \cdot 3$.
- 0 is even, because $0 = 2 \cdot 0$.
- -3 is odd, because $-3 = 2 \cdot (-2) + 1$.
- 7 is odd, because $7 = 2 \cdot 3 + 1$.
- 2.5 is neither even nor odd, because it is not an integer.

Definition 0.6 (Natural numbers (our convention))**.** In these notes, the **natural numbers** (Wikipedia contributors 2026c) are the positive integers (Definition 0.2):

$$\mathbb{N} \stackrel{\text{def}}{=} \{1, 2, 3, ...\}$$

Definition 0.7 (Non-negative integers)**.** The **non-negative integers** are the natural numbers (Definition 0.6) together with 0:

$$\mathbb{N}_0 \stackrel{\text{def}}{=} \{0, 1, 2, 3, ...\} = \mathbb{N} \cup \{0\}$$

Example 0.3 (Natural numbers in a data analysis)**.**

- Observation indices start at 1, so we write $i \in \{1, \dots, n\}$ with $n \in \mathbb{N}$.
- A count outcome, such as the number of hospital visits in a year, can be 0, so the set of values it can take is $\mathbb{N}_0$, not $\mathbb{N}$.

i Other conventions for the natural numbers

Other sources may define $\mathbb{N}$ differently, so check each source's definition before reading its formulas:

- Many sources include 0 in $\mathbb{N}$. The international standard ISO 80000-2 defines $\mathbb{N}$ to include 0, continuing the earlier standard ISO 31-11 (1978).
- Other sources start $\mathbb{N}$ at 1, as these notes do.
- To remove the ambiguity, some sources write $\mathbb{N}_1$ or $\mathbb{Z}^+$ for $\{1, 2, 3, ...\}$, and $\mathbb{N}_0$ or $\mathbb{Z}^{0+}$ for $\{0, 1, 2, ...\}$.
- The words vary too. "Positive integers" ($\{1, 2, 3, ...\}$) and "non-negative integers" ($\{0, 1, 2, ...\}$) are unambiguous. "Whole numbers" usually includes 0, but can also mean all of the integers, negative ones included. "Counting numbers" usually starts at 1, but some sources include 0.
- Some older texts write J for the natural numbers.

When a formula's meaning depends on whether 0 is included, we write the set out explicitly, for example $\{0, 1, 2, ...\}$ or $\{1, \dots, n\}$.
Source: Wikipedia, "Natural number", "Terminology and notation" and "Zero as natural number", which cites ISO 80000-2:2019 and the textbooks using each notation.

Rational, real, and irrational numbers

Definition 0.8 (Rational numbers)**.** A **rational number** (Wikipedia contributors 2026f) is a number that can be written as a fraction $\frac{a}{b}$, where a and b are integers (Definition 0.1) and $b \neq 0$. The set of rational numbers is written $\mathbb{Q}$:

$$\mathbb{Q} \stackrel{\text{def}}{=} \left\{ \frac{a}{b} : a \in \mathbb{Z},\ b \in \mathbb{Z},\ b \neq 0 \right\}$$

Example 0.4 (Rational numbers)**.**

- $\frac{3}{4}$ is rational, with $a = 3$ and $b = 4$.
- 0.75 is rational, because $0.75 = \frac{3}{4}$.

- Every integer is rational: for example, $-5 = \frac{-5}{1}$.
- $0.333\ldots$, with the 3s repeating forever, is rational, because $0.333\ldots = \frac{1}{3}$.

Definition 0.9 (Real numbers)**.** A **real number** (Wikipedia contributors 2026g) is a number that can be written as a decimal expansion: an integer part, followed by a decimal point and a finite or infinite list of decimal places after it. Equivalently, the real numbers are the points on the number line. The set of real numbers is written $\mathbb{R}$.

Remark (A working definition)

Remark 0.1 (A working definition)*.* Definition 0.9 is a working definition, not a construction: it does not say what an infinite decimal expansion is, or how to add or multiply two of them. Rudin (1976, Appendix to Chapter 1) constructs the real numbers rigorously. Some real numbers have two decimal expansions: for example, $0.999\ldots$ and $1.000\ldots$ are the same real number, 1.

Example 0.5 (Real numbers)**.**

- 3, -0.5, and $\frac{1}{3} = 0.333\ldots$ are real numbers.
- $\pi = 3.14159\ldots$ is a real number.
- Every rational number (Definition 0.8) is a real number, so $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.

Definition 0.10 (Irrational numbers)**.** An **irrational number** (Wikipedia contributors 2026b) is a real number (Definition 0.9) that is not rational (Definition 0.8).

Example 0.6 (Irrational numbers)**.**

- $\sqrt{2} = 1.41421\ldots$, the square root of 2, is irrational: no fraction of integers $\frac{a}{b}$ squares to 2 (Wikipedia contributors 2026h).
- $\pi = 3.14159\ldots$ is irrational (Wikipedia contributors 2026e).
- 1.5 is not irrational, because $1.5 = \frac{3}{2}$ is rational.

Further reading

- Wikipedia contributors (2026a) — the integers
- Wikipedia contributors (2026d) — even and odd numbers
- Wikipedia contributors (2026c) — the natural numbers and their conventions

- Wikipedia contributors (2026f) — the rational numbers
- Wikipedia contributors (2026g) — the real numbers and their constructions
- Wikipedia contributors (2026b) — irrational numbers
- Wikipedia contributors (2026h) — the square root of 2
- Wikipedia contributors (2026e) — proofs that pi is irrational

Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.

Wikipedia contributors. 2026a. *Integer — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Integer.

Wikipedia contributors. 2026b. *Irrational Number — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Irrational_number.

Wikipedia contributors. 2026c. *Natural Number — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Natural_number.

Wikipedia contributors. 2026d. *Parity (Mathematics) — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Parity_(mathematics).

Wikipedia contributors. 2026e. *Proof That Pi Is Irrational — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Proof_that_%CF%80_is_irrational.

Wikipedia contributors. 2026f. *Rational Number — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Rational_number.

Wikipedia contributors. 2026g. *Real Number — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Real_number.

Wikipedia contributors. 2026h. *Square Root of 2 — Wikipedia, the Free Encyclopedia*. https://en.wikipedia.org/wiki/Square_root_of_2.